

Student ID:	24101226	Lab Section:	11 A
Name:	Mehedi Hasan Rowdho	Lab Group:	06

Experiment-04

Study of Half-Wave and Full-Wave Rectifier

CSE251 - Electronic Devices and Circuits Lab



Objective

1. To build a half-wave rectifier circuit and understand its operating principle.
2. To build a full-wave rectifier circuit and understand its operating principle.

Equipments

1. p-n junction diode (1N4007) x 4
2. Resistor (10k Ω) - x1
3. Capacitors (1 μ F, 4.7 μ F) - 1 each
4. Function Generator
5. Oscilloscope
6. Multimeter
7. Breadboard and Wires

Background Theory

Diodes are used to build rectifier circuits which convert the input sinusoidal voltage V_s to a unipolar output V_o . There are two types of rectifier circuits: (i) Half-wave rectifier and (ii) Full-wave rectifier.

Half-Wave (HW) Rectifier

The circuit of a half-wave rectifier is shown in the following figure:

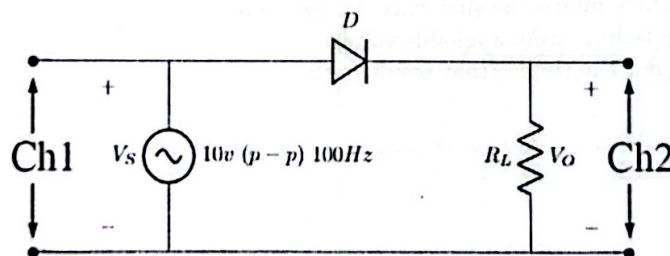


Figure 1: Half-Wave Rectifier

Assuming ideal diode model,

For the period, $t = 0 \rightarrow T/2$, $V_s > 0$, Diode is ON and $V_o = V_s$

For the period, $t = T/2 \rightarrow T$, $V_s < 0$, Diode is OFF and $V_o = 0$

As only positive half cycle appears at the output and the negative half is blocked, the AC input voltage changes into a unidirectional DC voltage at the output. The process of removing half of the input signal to establish a dc level is aptly called half-wave rectification. Due to diode voltage drop, the actual output voltage will be

approximately, $V_O = V_S - V_{D0}$. For, $V_S = V_m \sin \omega t$, DC voltage and current of a half wave rectifier are:

$$V_{dc} = \frac{V_m}{\pi} - \frac{V_{D0}}{2} \quad \text{and} \quad I_{dc} = \frac{V_{dc}}{R} = \frac{V_m/\pi - V_{D0}/2}{R}$$

where, V_m = peak input voltage, $V_{D0} \approx 0.7v$ for Silicon diodes. The following figures show the input and output wave shapes of a HW rectifier circuit:

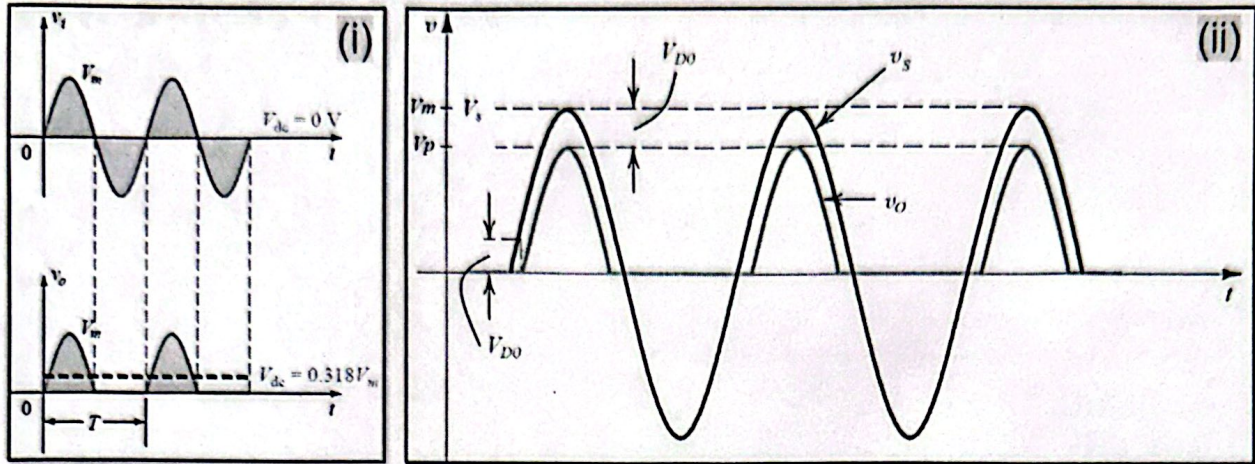


Figure 2: (i) Assuming Ideal Diode (ii) Assuming Real Diode (CVD Model)

HW Rectifier with Filter Capacitor

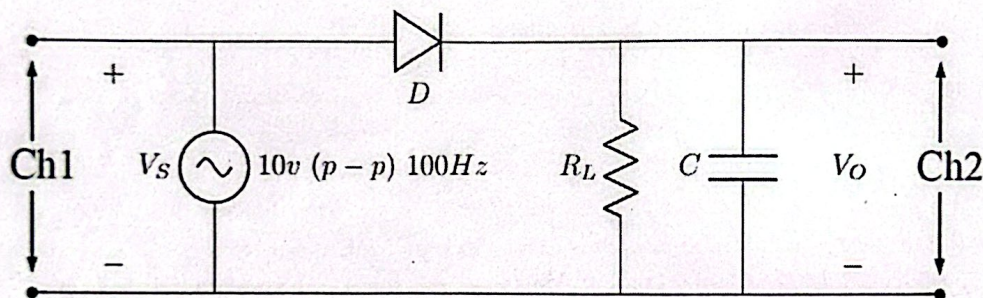


Figure 3: Half-Wave Rectifier with Filter Capacitor

Although the rectification stage makes the sine wave voltage to be positive, the rectifier's result is not as "flat" a DC value as we would like to have from a reliable voltage source, as you will measure in the lab. The capacitor is included to help smooth out the ripples that result in the output from the rectification stage.

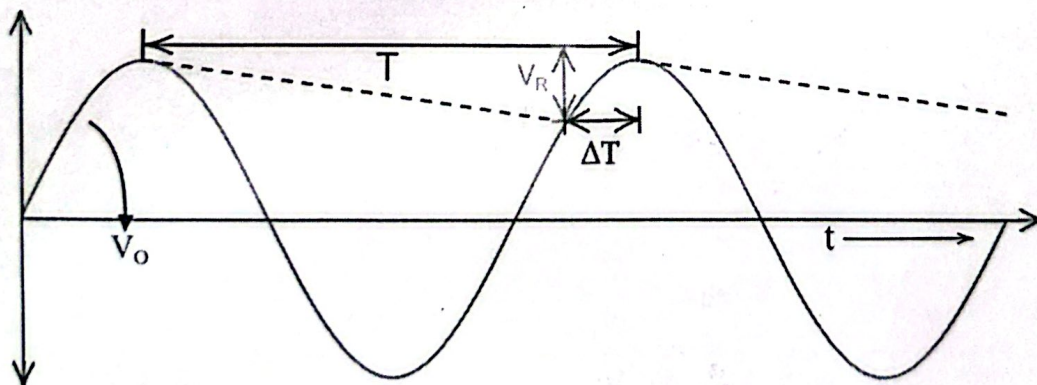


Figure 4: V_r or $V_{r(p-p)}$ is the peak to peak ripple voltage

Recall that the voltage across a capacitor cannot change instantaneously, but rather it requires a certain amount

of time before it is fully charged. Initially, as the input voltage rises, diode turns on, and the capacitor starts charging. After the input voltage reaches its peak value, the capacitor gets charged to the peak input voltage. As the input voltage now starts decreasing below the peak value, the diode turns off, since the n-side is more positive than the p-side due to the voltage across the capacitor which does not change instantaneously. The stored charges on the capacitor will be released through R_L .

For $R_L C \gg T$, it will take long time for the capacitor to discharge and the output terminal will maintain almost a dc voltage. Thus, large capacitance values help suppress the quickly changing voltage from the rectifier and result in a flatter DC value being supplied to the load. Typical power supply designs use relatively large capacitor values (greater than 1000 μF).

Peak Inverse Voltage(PIV): PIV is the maximum voltage that appears across the diode when it is reverse-biased.

$$PIV = V_m$$

Ripple Voltage and Ripple Factor

The output of a rectifier though unidirectional, contains periodically fluctuating components. The theoretical value for the peak-to-peak ripple voltage is given by,

$$V_{r(p-p)} = \frac{V_p}{fCR_L} = \frac{V_m - V_{D0}}{fCR_L}$$

Here, V_p = peak voltage of the rectified output, f = input frequency, and R_L is the resistance connected in parallel with C .

$$\text{Average value of the output wave, } V_{dc} = V_p - \frac{V_{r(p-p)}}{2}$$

$$\text{RMS value of the Ripple Voltage, } V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

A measure of the fluctuating components is given by the ripple factor r , which is defined as,

$$r = \frac{\text{rms value of alternating components of the output wave (multimeter in AC)}}{\text{average value of the output wave (multimeter in DC)}}$$

$$= \frac{V_{r-rms}}{V_{dc}}$$

Full-Wave (FW) Rectifier / Bridge Rectifier

The full-wave rectifier utilizes both halves of the input sinusoid. To provide a unipolar output, it inverts the negative halves of the sine wave. Figure-5 shows the circuit diagram, input and output waveform of Full-Wave Rectifier/Bridge Rectifier.

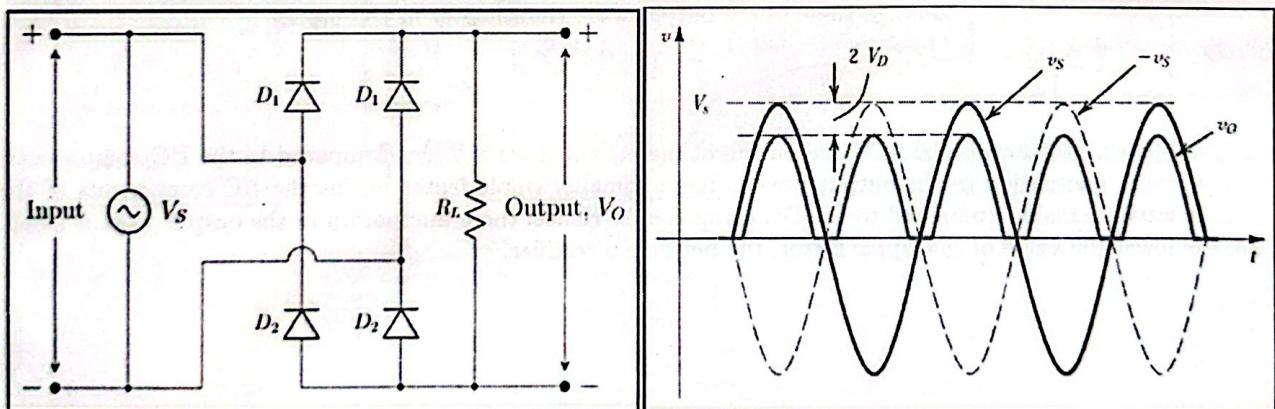


Figure 5: *Full-Wave Rectifier Circuit without Capacitor and the Waveform of Input and Output*

In this case, constant voltage drop model was assumed. Peak inverse voltage across each diode and DC voltage in reverse-bias can be calculated using the following equations,

$$PIV = V_m - V_{D0}$$

$$V_{dc} = \frac{2V_m}{\pi} - 2V_{D0}$$

Full-Wave (FW) Rectifier with Capacitor

The pulsating nature of the output voltage produced by the rectifier circuits discussed above makes it unsuitable as a dc supply for electronic circuits. A simple way to reduce the variation of the output voltage is to place a capacitor across the load resistor. Figure-6 shows the circuit diagram and output waveform of Full-Wave Rectifier with Capacitor.

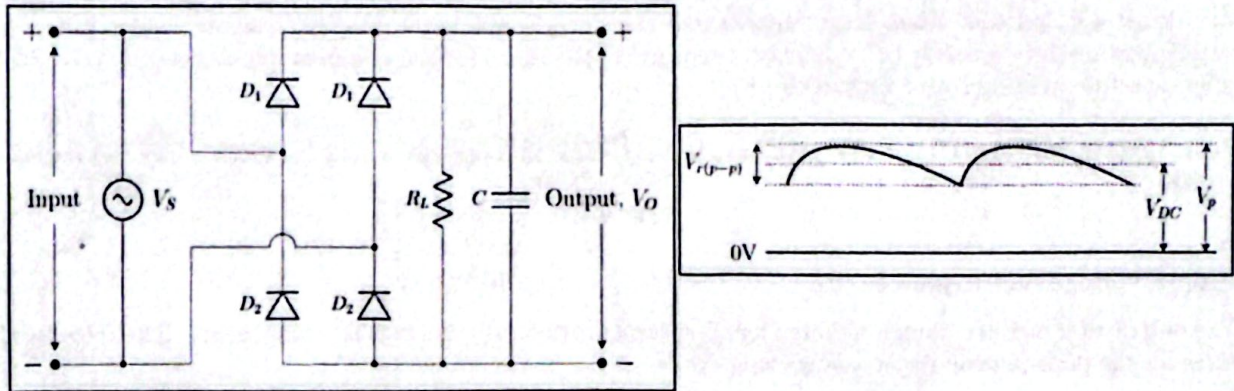


Figure 6: Full-Wave Rectifier Circuit with Capacitor and the Output Waveform

Ripple Voltage and Ripple Factor

The output of a rectifier though unidirectional, contains periodically fluctuating components. The theoretical value for the peak-to-peak ripple voltage is given by,

$$V_{r(p-p)} = \frac{V_p}{2fCR_L} = \frac{V_m - 2V_{D0}}{2fCR_L}, \text{ for a FW rectifier circuit}$$

Here, V_p = peak voltage of the rectified output, f = input frequency, and R is the resistance connected in parallel with C .

$$\text{Average value of the output wave, } V_{dc} = V_p - \frac{V_{r(p-p)}}{2}$$

$$\text{RMS value of the Ripple Voltage, } V_{r-rms} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

A measure of the fluctuating components is given by the ripple factor r , which is defined as

$$r = \frac{\text{rms value of alternating components of the output wave (multimeter in AC mode)}}{\text{average value of the output wave (multimeter in DC mode)}}$$

$$= \frac{V_{r-rms}}{V_{dc}}$$

Larger ripple factor means the AC components of the output wave is larger compared to the DC components. Hence, there fluctuation in the output wave is large. Smaller ripple factor means the AC components of the output wave is smaller compared to the DC components. Hence, there fluctuation in the output wave is small. So, the lower the value of the ripple factor, the better the rectifier.

Task 01: HW Rectifier

Procedure:

1. Construct the circuit of Figure 7 with $R_L = 10\text{ k}\Omega$ and observe V_s and V_o simultaneously on the oscilloscope using Channel 1 and Channel 2 simultaneously.
2. Capture the image of the input and output waveforms simultaneously using mobile camera.
3. Use *Measure* of the oscilloscope to measure the peak voltage of the input and output waveforms, and also the peak-to-peak ripple voltage, $V_{r(p-p)}$.
4. Again, connect the multimeter across the resistor and measure the ac and dc value of the output voltage by using *ac* and *dc* mode of the multimeter in *voltage* setting respectively.
5. Connect $1\text{ }\mu\text{F}$ capacitor parallel to the resistor, R_L and repeat steps 1-4.
6. Replace the capacitor with a $4.7\text{ }\mu\text{F}$ capacitor and repeat steps 1-4 again.

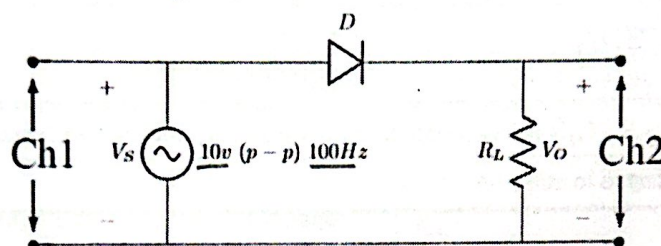


Figure 7: Half-Wave Rectifier

HW Rectifier Experimental Observation

Capacitor	Measure (using oscilloscope)			Measure (using multimeter)		Calculate	
	Peak Input Voltage, V_m	Peak Output Voltage, V_p	Peak-to- peak ripple voltage, $V_{r(p-p)}$	Average/DC value of output voltage, V_{dc}	AC value of output voltage, V_{ac}	Diode voltage, $V_{D0} = V_m - V_p$	Ripple factor, $r = \frac{V_{ac}}{V_{dc}}$
None	5.04	4.48	4.48	1.356	1.706	0.56	1.258
$1\text{ }\mu\text{F}$		4.4	2.46	2.850	0.741	0.64	0.26
$4.7\text{ }\mu\text{F}$		4.32	0.96	3.88	0.226	0.72	0.058

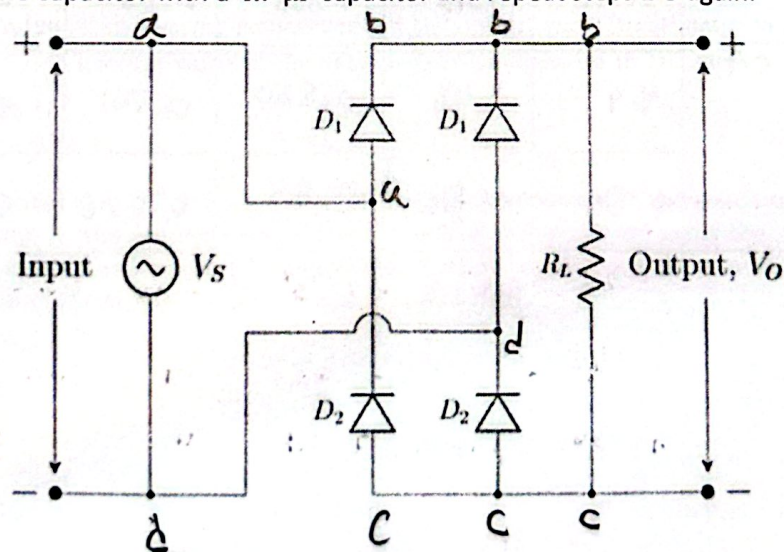
HW Rectifier Theoretical Calculations

Calculate the following using the experimental values of V_m , V_p , $V_{r(p-p)}$, and V_{D0}			
Capacitor	Average/DC value of output voltage, $V_{dc} = \begin{cases} \frac{V_m}{\pi} - \frac{V_{D0}}{2}, & \text{no Capacitor} \\ V_p - \frac{V_{r(p-p)}}{2}, & \text{with Capacitor} \end{cases}$	AC value of output voltage, $V_{ac} = \begin{cases} \frac{V_p}{2}, & \text{no Capacitor} \\ \frac{V_{r(p-p)}}{2\sqrt{3}}, & \text{with Capacitor} \end{cases}$	Ripple factor, $r = \frac{V_{ac}}{V_{dc}}$
None	1.32	2.42	1.83
1 μF	3.14	0.71	0.23
4.7 μF	3.84	0.28	0.073

Task 02: FW Rectifier

Procedure:

- Construct the circuit of Figure 11 with $R_L = 10 \text{ k}\Omega$ and observe V_s (10V p-p , 100 Hz) using Channel 1 of the oscilloscope and capture the image of the input waveform.
- Measure the maximum value of the input voltage using *Measure* of the oscilloscope.
- Disconnect the channel and connect it across the output to observe V_o . Capture the image of the output waveform using mobile camera.
- Use *Measure* of the oscilloscope to measure the peak voltage of the output waveforms, and also the peak-to-peak ripple voltage, $V_{r(p-p)}$.
- Now, connect the multimeter across the resistor and measure the ac and dc value of the output voltage by using *ac* and *dc* mode of the multimeter in *voltage* setting respectively.
- Connect $1 \mu\text{F}$ capacitor parallel to the resistor, R_L and repeat steps 1-5.
- Replace the capacitor with a $4.7 \mu\text{F}$ capacitor and repeat steps 1-5 again.



FW Rectifier Experimental Observation

	Measure (using oscilloscope)			Measure (using multimeter)		Calculate	
Capacitor	Peak Input Voltage, V_m	Peak Output Voltage, V_p	Peak-to- peak ripple voltage, $V_{r(p-p)}$	Average/DC value of output voltage, V_{dc}	AC value of output voltage, V_{ac}	Diode voltage, $V_{D0} = \frac{V_m - V_p}{2}$	Ripple factor, $r = \frac{V_{ac}}{V_{dc}}$
None	5.08	3.92	4.04	2.16	1.320	0.58	0.61
1 μ F		3.80	1.42	3.012	0.396	0.64	0.131
4.7 μ F		3.46	0.54	3.476	0.104	0.81	0.03

FW Rectifier Theoretical Calculations

	Calculate the following using the experimental values of V_m , V_p , $V_{r(p-p)}$, and V_{D0}		
Capacitor	Average/DC value of output voltage, $V_{dc} = \begin{cases} \frac{2 V_m}{\pi} - 2 V_{D0}, & \text{no Capacitor} \\ V_p - \frac{V_{r(p-p)}}{2}, & \text{with Capacitor} \end{cases}$	AC value of output voltage, $V_{ac} = \begin{cases} \sqrt{\left(\frac{V_m}{\sqrt{2}}\right)^2 - V_{dc}^2}, & \text{no Capacitor} \\ \frac{V_{r(p-p)}}{2\sqrt{3}}, & \text{with Capacitor} \end{cases}$	Ripple factor, $r = \frac{V_{ac}}{V_{dc}}$
None	2.074	2.87	1.38
1 μ F	3.09	0.41	0.133
4.7 μ F	3.19	0.16	0.05

Table for Comparison

		Experimental Observations			Theoretical Calculation		
Rectifier	C (μ F)	V_{ac} (V)	V_{dc} (V)	r	V_{ac} (V)	V_{dc} (V)	r
HW	1	0.741	2.85	0.26	0.71	3.14	0.23
	4.7	0.226	3.88	0.058	0.28	3.84	0.073
FW	1	0.396	3.012	0.131	0.41	3.09	0.133
	4.7	0.104	3.476	0.03	0.16	3.19	0.05

Signature of the lab faculty

Zayed Anju
13/4/26

Report Submission Guidelines

1. Add a cover page that includes course code, course title, name, student ID, group, semester, date of performance, date of submission.
2. Attach the captured photos of all the waveforms and measurements you have observed in the Oscilloscope. Each photo should contain necessary description.
3. Answer the questions in the "Test Your Understanding" section.
4. Add a brief Discussion regarding the experiment. For the Discussion part of the lab report, you should include the answers of the following questions in your own words:
 - What did you learn from this experiment?
 - What challenges did you face and how did you overcome the challenges? (if any)
 - What mistakes did you make and how did you correct the mistakes? (if any)
 - How will this experiment help you in future experiments of this course?

Test Your Understanding

Answer the following questions:

1. Which capacitor acts as a better filter? Explain briefly.

Answer:

4.7 μF capacitor is better filter. Capacitor with higher capacity can hold more charge and it also discharges slowly. We know,

$$\text{For discharge, } V(t) = V(0)e^{-t/RC} \quad \text{--- ①}$$

From ① we can see that the higher the value of RC gets the slower the rate of discharge gets. If we keep R the same and increase the value of C then it will discharge slower.

So, 4.7 μF capacitor is better than 1 μF capacitor.

2. Which of the two rectifiers is better? Which quantity is calculated for this purpose? Explain briefly.

Answer:

A full wave rectifier is better than a half wave rectifier. A FW-rectifier uses both halves of the AC signal but a HW-rectifier only use one half cycle. As FW uses both half cycles it gives better, smoother and more efficient DC output.

The main quantity used to compare performance is ripple factor.

Which is the measurement of fluctuation. $r = \frac{V_{AC}}{V_{DC}}$

Less ripple factor means less fluctuation and better performance. FW gives lesser ripple factors than HW.

3. Why can't you see the input and output using both channels of the oscilloscope simultaneously in Task-02?

Answer:

In the oscilloscope the negative terminal of the both channels are internally short. That's why while measuring we have to connect them at the same node or just connect only one of them. But in F-W rectifier circuit negative terminal of the input and output signal are on different node. If we connect output and input both end together with the oscilloscope it would create a short circuit between those two different nodes or cause disturbance. Which will not allow to display both signal correctly.

Discussion

From this experiment I learnt how both Half wave rectifier and Full wave rectifier behave in practical. I observed how the characteristics of the graphs change by not using any capacitor, by using $1\mu\text{F}$ capacitor and by using $4.7\mu\text{F}$ capacitor. I learn that the higher the capacitance gets the better the result gets. I also learnt that FW gives better performance than HW. For the task01 constructing the

circuit was easy. However, while constructing the circuit of the task02 I was facing difficulties. I was constructing it vertically which made difficult. But then I realize constructing the circuit horizontally would make it easier. Also, I was facing difficulties in understanding the positive and negative terminals of the diodes at first. But later I figured that out. The mistake I made was placing the diodes by the wrong pole in task01. But after checking with multimeter I corrected that mistake. This experiment made my basic knowledge for rectifier clearer. Which will help me to understand and solve rectifier related complex problems in the upcoming experiments of this course.

GWINSTEK

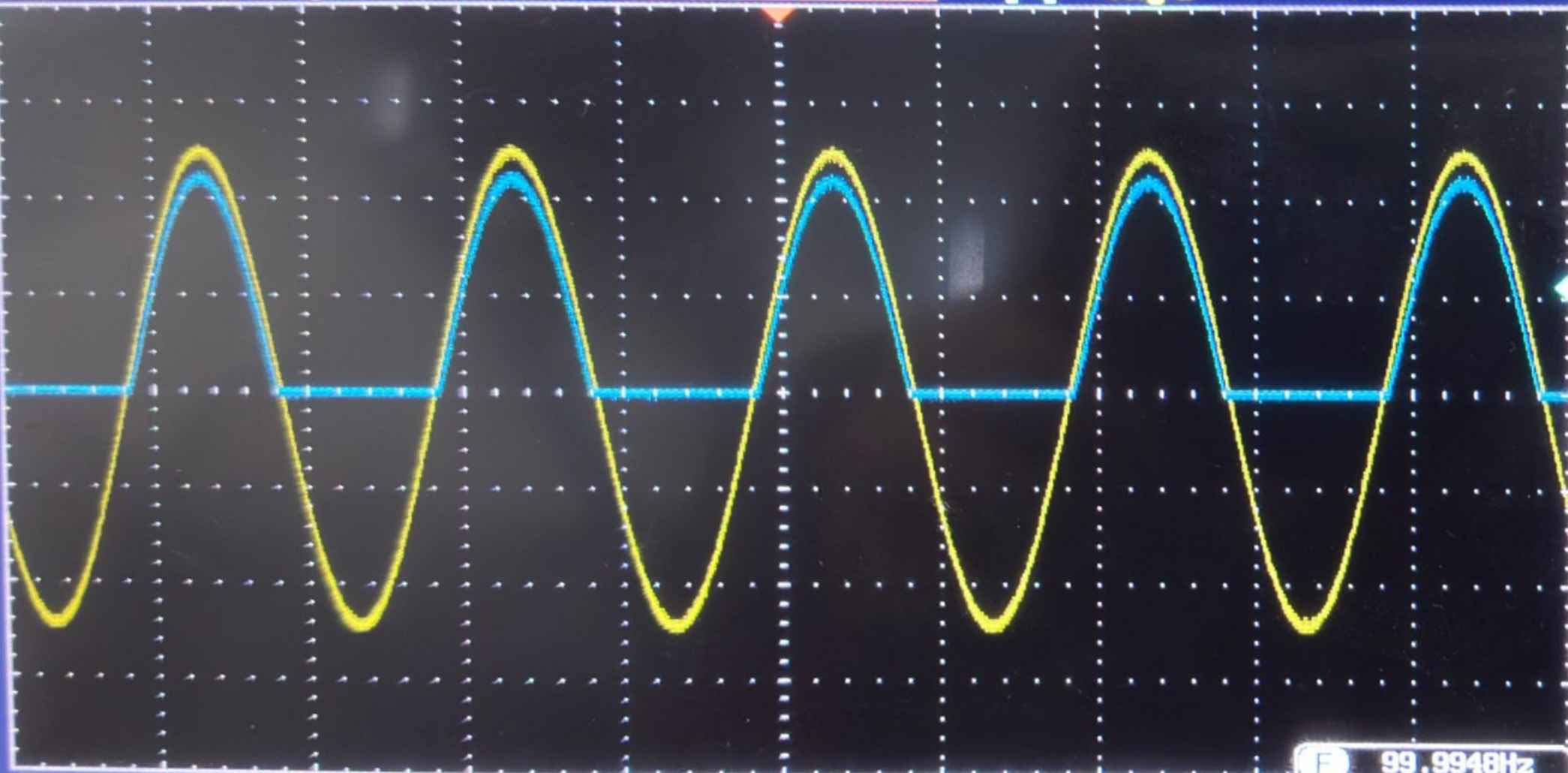
10k pts

200kSa/s



Trig'd

2



① $\approx 2.00V$ ② $\approx 2.00V$

5ns

0.00000s

F 99.9948Hz

② f 2.24V DC

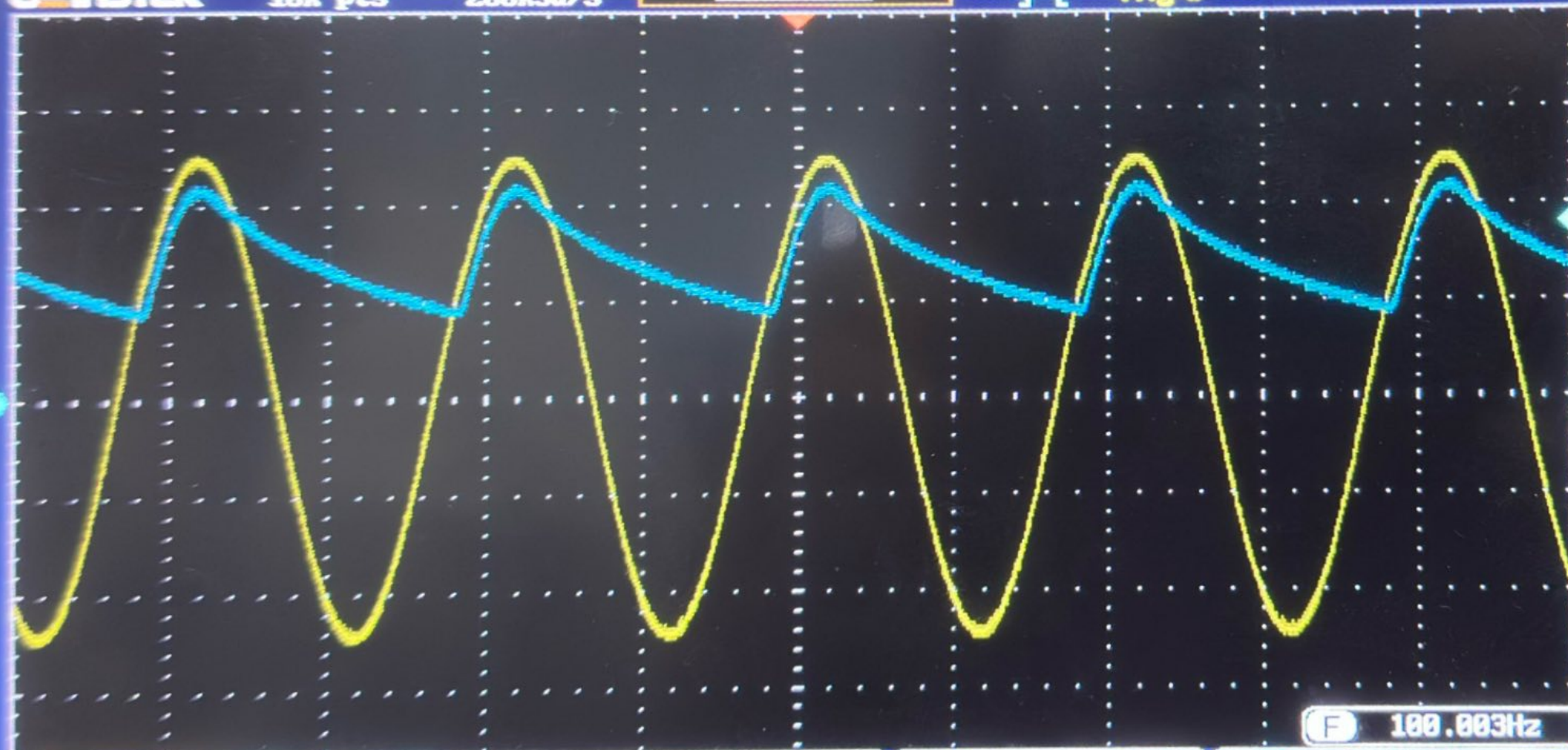
GWINSTEK

10k pts

200kSa/s



Trig'd



1 = 2.00V 2 = 2.00V

5ns

0.00000s

2

f

3.6V

DC

Mode
Fit Screen
AC Priority

Undo
Autoset

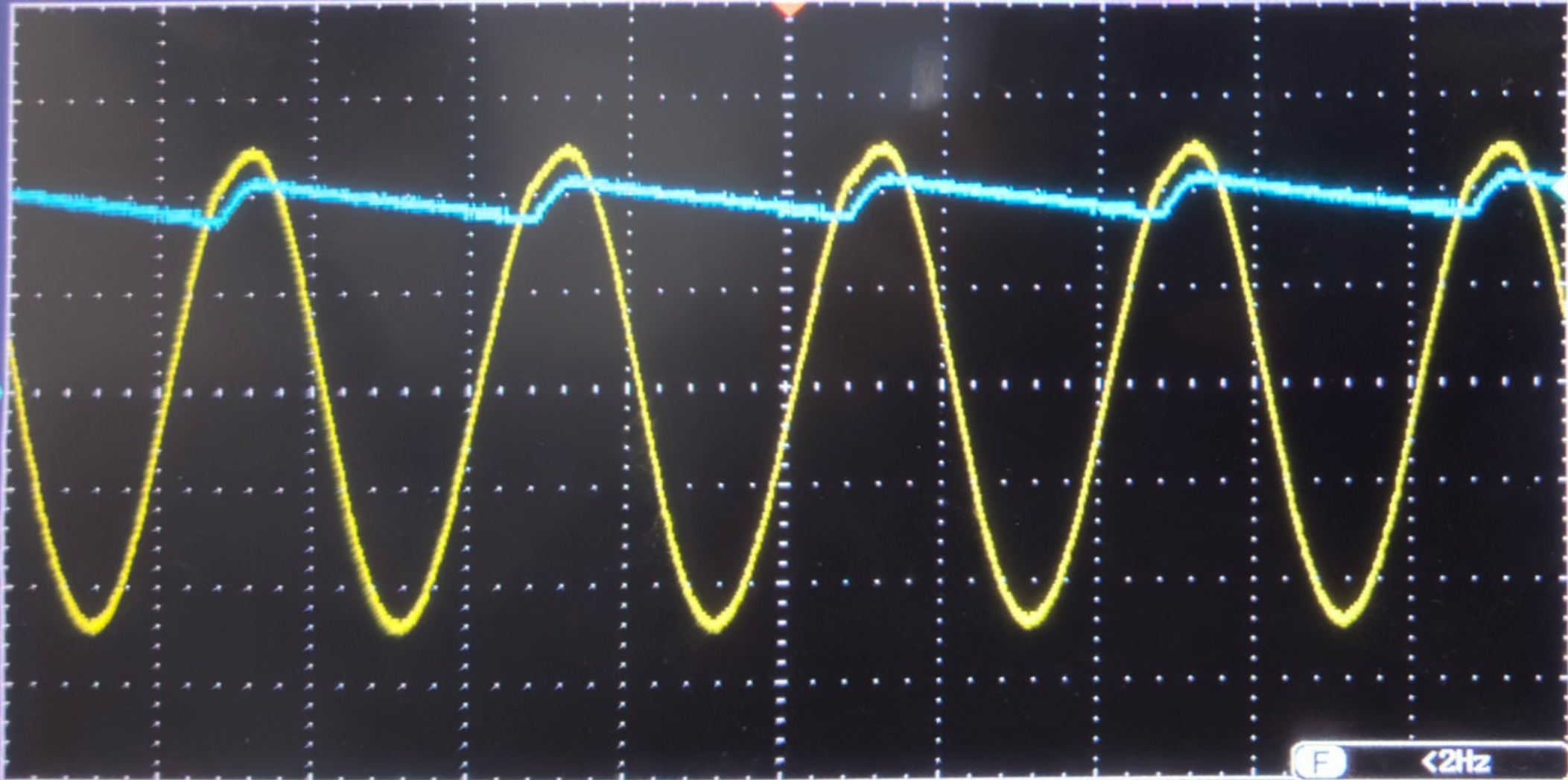
GWINSTEK

10k pts

200kSa/s



Stop



① 2.00V ② 2.00V

5ns 0.00000s

② f 4V DC

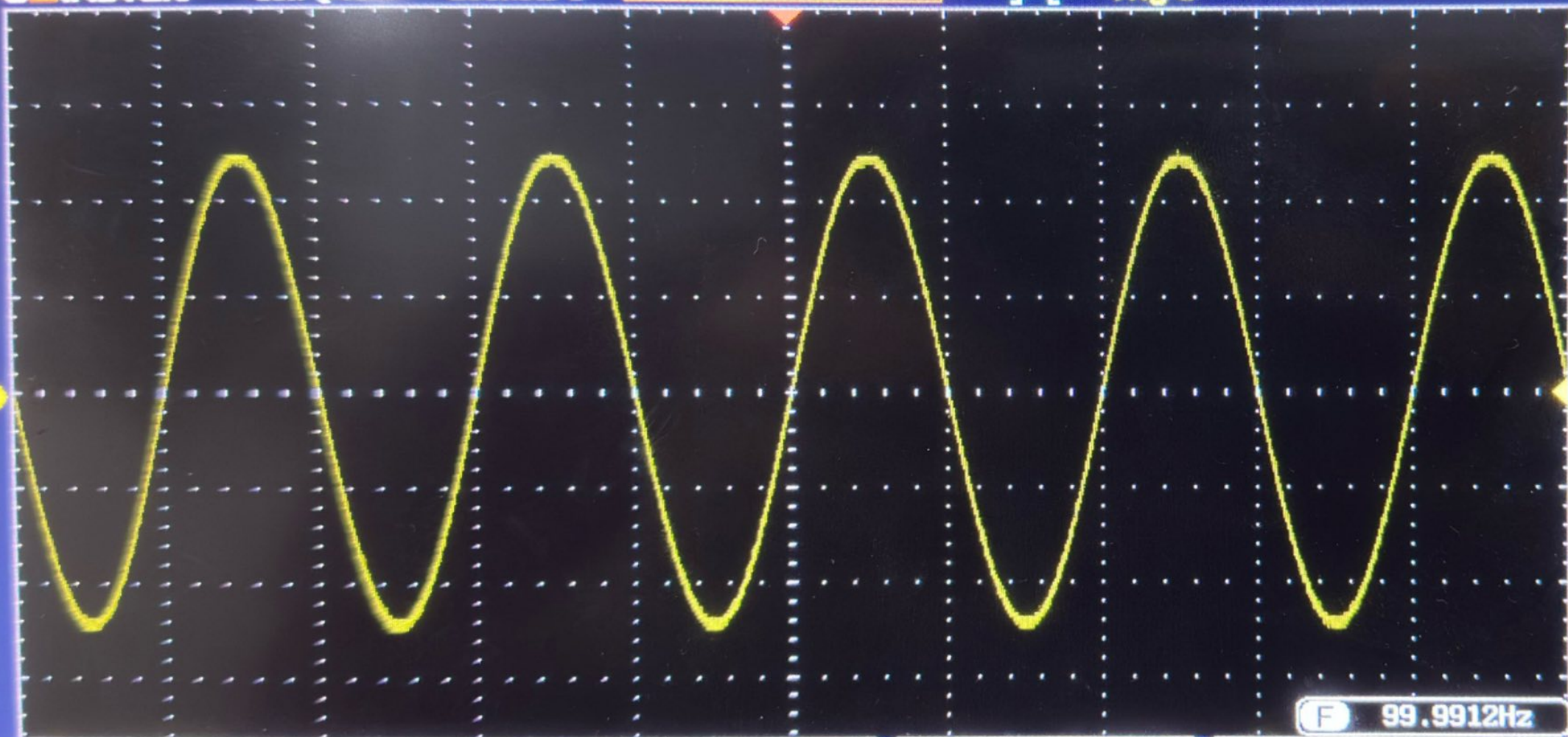
GWINSTEK

10k pts

200kSa/s



Trig'd



1 = 2.00V 2 = 50.0mV

5ns

0.00000s

F 99.9912Hz

1 f 80mV

DC

Mode
Fit Screen
AC Priority

Undo
Autoset

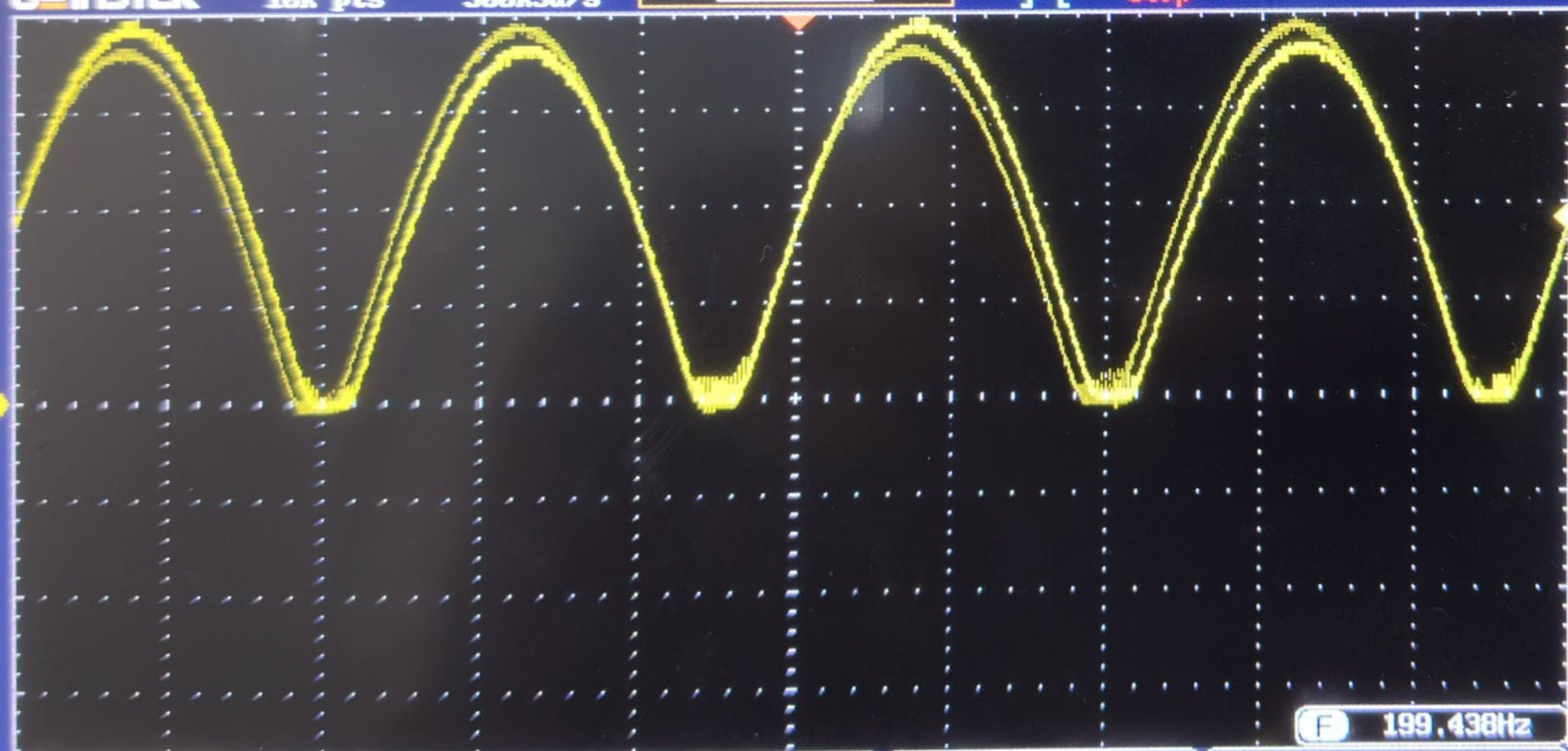
GWINSTEK

10k pts

500kSa/s



Stop



1 = 1.00V 2 = 50.0mV

2ns

0.00000s

F 199.438Hz

1 1.84V DC

Mode
Fit Screen
AC Priority

Undo
Autoset

GW15TEK

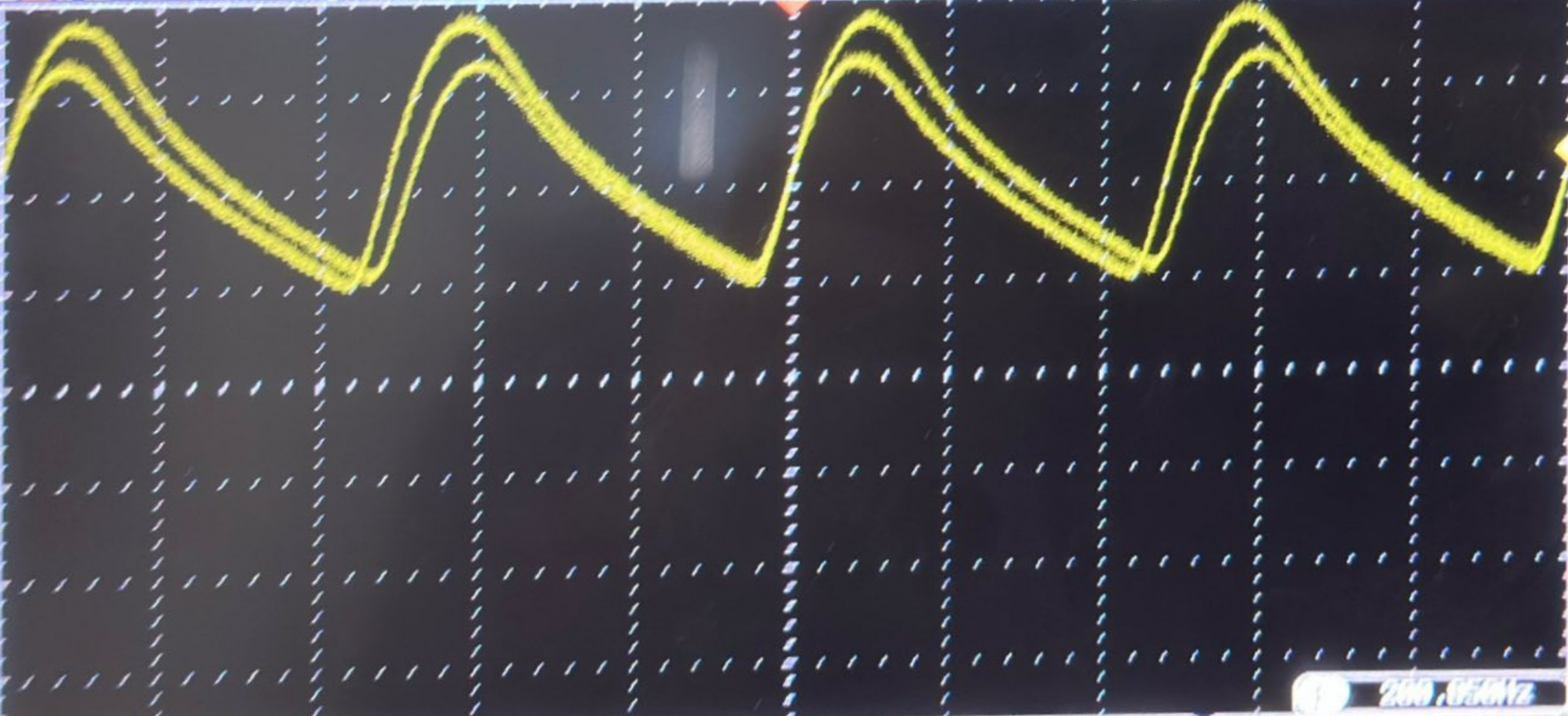
10k pts

500kSa/s

0.000000

Trig'd

200.0561Hz



2ms

0.000000

1

3.020

DC

Mode
Fit Screen
AC Priority

Undo
Autoset

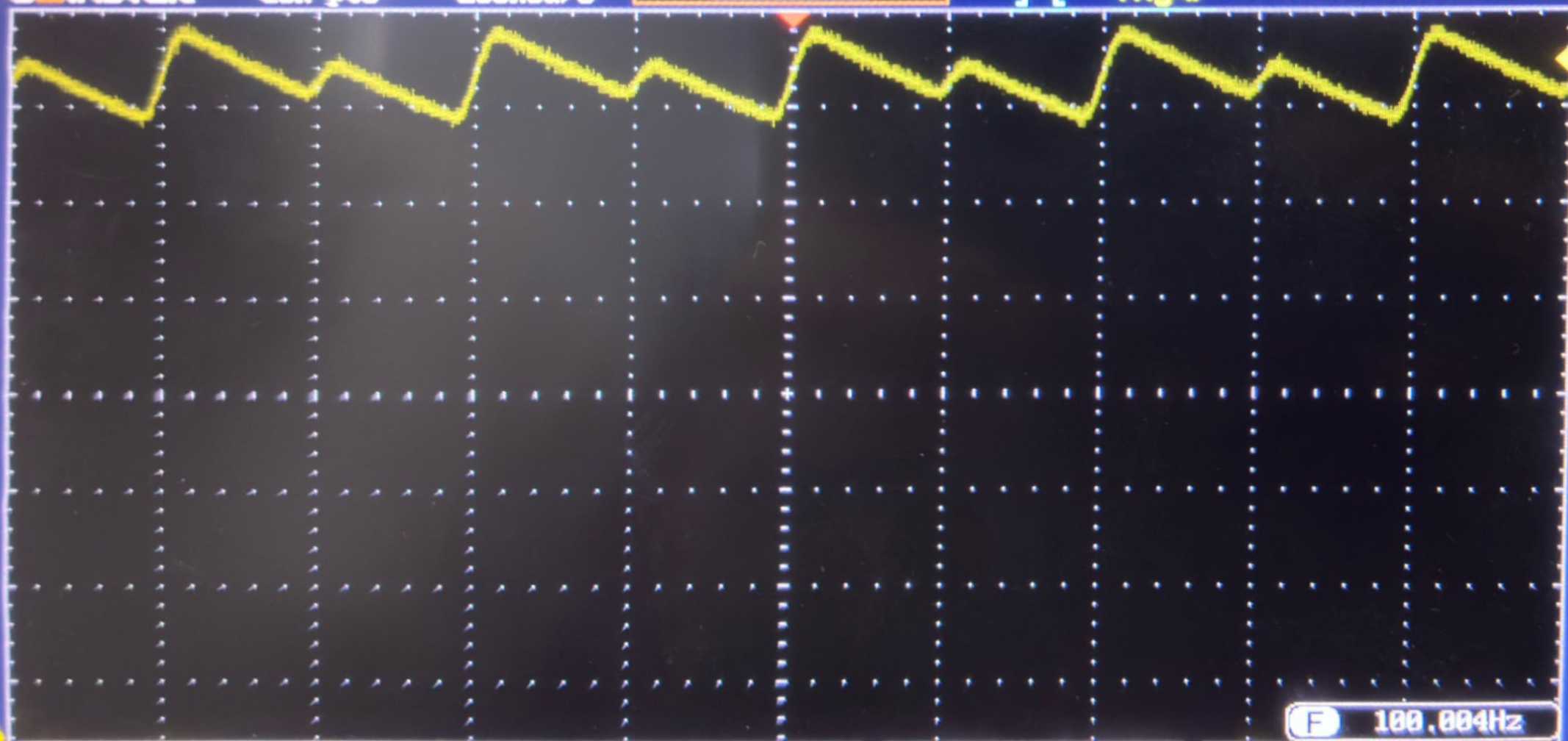
GWINSTEK

10k pts

200kSa/s



Trig'd



① = 500mV ② = 50.0mV

5ms

0.00000s

F 100.004Hz

① f 3.54V DC

Mode
Fit Screen
AC Priority

Undo
Autoset